

```
PS C:\Users\JAY> mkdir serverless-url-shortener
```

```
Directory: C:\Users\JAY
```

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	21-04-2026 15:05		serverless-url-shortener

```
PS C:\Users\JAY\serverless-url-shortener> mkdir lambda, docker, terraform
```

Directory: C:\Users\JAY\serverless-url-shortener

Mode	LastWriteTime		Length	Name
----	-----		-----	----
d-----	21-04-2026	15:06		lambda
d-----	21-04-2026	15:06		docker
d-----	21-04-2026	15:06		terraform

```
PS C:\Users\JAY\serverless-url-shortener> New-Item -ItemType File -Path README.md, deploy.sh, destroy.sh
```

```
Directory: C:\Users\JAY\serverless-url-shortener
```

Mode	LastWriteTime		Length	Name
----	-----		-----	----
-a----	21-04-2026	15:07	0	README.md
-a----	21-04-2026	15:07	0	deploy.sh
-a----	21-04-2026	15:07	0	destroy.sh

```
PS C:\Users\JAY\serverless-url-shortener> New-Item -ItemType File -Path lambda/create_url.py, lambda/redirect_url.py, lambda/requirements.txt
```

Directory: C:\Users\JAY\serverless-url-shortener\lambda

Mode	LastWriteTime		Length	Name
----	-----		-----	----
-a----	21-04-2026	15:07	0	create_url.py
-a----	21-04-2026	15:07	0	redirect_url.py
-a----	21-04-2026	15:07	0	requirements.txt

```
PS C:\Users\JAY\serverless-url-shortener> New-Item -ItemType File -Path docker/Dockerfile, docker/.dockerignore
```

```
Directory: C:\Users\JAY\serverless-url-shortener\docker
```

Mode	LastWriteTime		Length	Name
----	-----		-----	----
-a----	21-04-2026	15:07	0	Dockerfile
-a----	21-04-2026	15:07	0	.dockerignore

```
PS C:\Users\JAY\serverless-url-shortener> New-Item -ItemType File -Path terraform/main.tf, terraform/variables.tf, terraform/outputs.tf
```

```
Directory: C:\Users\JAY\serverless-url-shortener\terraform
```

Mode	LastWriteTime		Length	Name
----	-----		-----	----
-a----	21-04-2026	15:30	0	main.tf
-a----	21-04-2026	15:30	0	variables.tf
-a----	21-04-2026	15:30	0	outputs.tf

```
jay@LAPTOP-HUF2I1SF:/mnt/c/Users/JAY/serverless-url-shortener$ terraform --version  
Terraform v1.14.9  
on linux_amd64
```



HashiCorp HCL

HashiCorp  hashicorp.com

 607,779

     (8)

HashiCorp HCL syntax

Disable 

Uninstall 

 Auto Update 

DETAILS

FEATURES

CHANGELOG

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

 wsl     ...  

jay@LAPTOP-HUF2I1SF:/mnt/c/Users/JAY/serverless-url-shortener\$

```
jay@LAPTOP-HUF2I1SF:/mnt/c/Users/JAY/serverless-url-shortener$ docker --version
```

```
Docker version 28.3.0, build 38b7060
```

```
jay@LAPTOP-HUF2I1SF:/mnt/c/Users/JAY/serverless-url-shortener$ python3 -m pip show boto3
Name: boto3
Version: 1.34.46
Summary: The AWS SDK for Python
Home-page: https://github.com/boto/boto3
Author: Amazon Web Services
Author-email:
License: Apache License 2.0
Location: /usr/lib/python3/dist-packages
Requires:
Required-by:
```

AWS Lambda

Run code without managing servers.

Get started

Create a function

Run

.NET	Java	Node.js	Python	Ruby	Custom runtime
------	------	---------	--------	------	----------------

Create function [Info](#)

Choose one of the following options to create your function.

- ☒ **Author from scratch**
Start with a simple Hello World example.
- ☐ **Use a blueprint**
Build a Lambda application from sample code and configuration presets for common use cases.
- ☐ **Container image**
Select a container image to deploy for your function.

Basic information

Function name

Enter a name that describes the purpose of your function.

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime [Info](#)

Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.12



Permissions

By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. To use a different role, choose **Custom execution** role under **More settings**.

Info [Tutorials](#) >

Learn how to implement common use cases in AWS Lambda.

Create a simple web app ^

In this tutorial you will learn how to:

- Build a simple web app, consisting of a Lambda function with a function URL that outputs a webpage
- Invoke your function through its function URL

[Learn more](#) 

[Start tutorial](#)

Functions (1)

Last fetched 27/04/2026, 15:17:40



Actions

Create function

Info

Tutorials



Learn how to implement common use cases in AWS Lambda.

Create a simple web app
In this tutorial you will learn how to:

Q Search by attributes or search by keyword

☐	Function name	Description	Package type	Runtime	Type	Last modified
☐	MY-LAMBDA-CREATE-URL	-	Zip	Python 3.12	Standard	13 seconds ago

Choose one of the following options to create your function.

- ☒

Author from scratch
Start with a simple Hello World example.
- ☐

Use a blueprint
Build a Lambda application from sample code and configuration presets for common use cases.
- ☐

Container image
Select a container image to deploy for your function.

Basic information

Function name

Enter a name that describes the purpose of your function.

MY-LAMBDA-REDIRECT-URL

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime

[Info](#)

Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.12



Permissions

By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. To use a different role, choose **Custom execution** role under **More settings**.

Learn how to implement common use cases in AWS Lambda.

Create a simple web app

In this tutorial you will learn how to:

- Build a simple web app, consisting of a Lambda function with a function URL that outputs a webpage
- Invoke your function through its function URL

[Learn more](#)

Start tutorial

Functions (2)

Last fetched 27/04/2026, 15:19:29 

Actions ▼

Create function

 *Search by attributes or search by keyword*

☐	Function name ▼	Description ▼	Package type ▼	Runtime ▼	Type ▼	Last modified
☐	MY-LAMBDA-CREATE-URL	-	Zip	Python 3.12	Standard	2 minutes ago
☐	MY-LAMBDA-REDIRECT-URL	-	Zip	Python 3.12	Standard	8 seconds ago

API Gateway

APIs

Custom domain names

Domain name access associations

VPC links

AgentCore targets [New](#)

▼ API: JAY-URL-SHORTNER (hb0gz7cirh)

Resources

Stages

Authorizers

Gateway responses

Models

Resource policy

API Gateway > APIs > Resources - JAY-URL-SHORTNER (hb0gz7cirh)

Successfully created REST API 'JAY-URL-SHORTNER (hb0gz7cirh)'.

API actions

Deploy API

Resources

Create resource

/

Resource details

Path /

Resource ID oac5hhivw9

Update documentation

Enable CORS

Methods (0)

Delete

Create method

Method type

Integration type

Authorization

API key

No methods

No methods defined.

Lambda function

Provide the Lambda function name or alias. You can also provide an ARN from another account.

us-east-1

▼

arn:aws:lambda:us-east-1:486264854158:function:MY-LAMBDA-CRE

✕

Grant API Gateway permission to invoke your Lambda function
When you save your changes, API Gateway updates your Lambda function's resource-based policy to allow this API to invoke it.

Integration timeout | [Info](#)

By default, you can enter an integration timeout of 50 - 29,000 milliseconds. You can use Service Quotas to raise the integration timeout to greater than 29,000 ms

29000

► **Method request settings**

► **URL query string parameters**

► **HTTP request headers**

► **Request body**

[Cancel](#)

Create method

Create resource

Resource details

☒ Proxy resource [Info](#)
Proxy resources handle requests to all sub-resources. To create a proxy resource use a path parameter that ends with a plus sign, for example {proxy+}.

Resource path

/

▼

Resource name

shortcode

☐ CORS (Cross Origin Resource Sharing) [Info](#)
Create an OPTIONS method that allows all origins, all methods, and several common headers.

Cancel

Create resource

Create resource

☐ /

POST

/shortcode

Resource details

Delete

Update documentation

Enable CORS

Path
/shortcode

Resource ID
lsowqv

Methods (0)

Delete

Create method

Method type ▲	Integration type ▼	Authorization ▼	API key ▼
No methods			
No methods defined.			

Method type

GET

Integration type

☒ Lambda function

Integrate your API with a Lambda function.



☐ HTTP

Integrate with an existing HTTP endpoint.



☐ Mock

Generate a response based on API Gateway mappings and transformations.



☐ AWS service

Integrate with an AWS Service.



☐ VPC link

Integrate with a resource that isn't accessible over the public internet.



☒ Lambda proxy integration

Send the request to your Lambda function as a structured event.

Response transfer mode [Info](#)

☒ Buffered

Wait to receive the complete response before beginning transmission.

☐ Stream

Send portions of the response without waiting for the complete response.

Lambda function

Provide the Lambda function name or alias. You can also provide an ARN from another account.

us-east-1

arn:aws:lambda:us-east-1:486264854158:function:MY-LAMBDA-REDI

Deploy API



Create or select a stage where your API will be deployed. You can use the deployment history to revert or change the active deployment for a stage. [Learn more](#) 

Stage

New stage



Stage name

prod



A new stage will be created with the default settings. Edit your stage settings on the **Stage** page.

Deployment description



[Cancel](#)

Deploy

✚ prod

Stage details [Info](#)

Edit

Stage name
prod

Rate [Info](#)
10000

Web ACL
-

Cache cluster [Info](#)
⊖ Inactive

Burst [Info](#)
5000

Client certificate
-

Default method-level caching
⊖ Inactive

Invoke URL
 https://hb0gz7cirh.execute-api.us-east-1.amazonaws.com/prod

Active deployment
ocati6 on April 27, 2026, 15:37 (UTC+05:30)

outputs.tfvariables.tfmaintfredirect_url.pycreate_url.pyXDockerrfi

lambda > create_url.py > lambda_handler

```
17 lambda_handler(event, context):
41     click_count = 0,
42     }
43 )
44
45 domain = os.environ.get(
46     "DOMAIN", "https://hb0gz7cirh.execute-api.us-east-1.amazonaws.com/prod"
47 )
48 short_url = f"{domain}/{short_code}"
49
50 return response(
51     200,
52     {"short_url": short_url, "short_code": short_code, "long_url": long_url},
53 )
54
55 except Exception as e:
56     return response(500, {"error": str(e)})
57
58
59 response(status_code, body):
```

Follow link (ctrl + click)

```
jay@LAPTOP-HUF2I1SF:/mnt/c/Users/JAY/serverless-url-shortener$ sudo chown root:docker /var/run/docker.sock
[sudo] password for jay:
jay@LAPTOP-HUF2I1SF:/mnt/c/Users/JAY/serverless-url-shortener$ sudo chmod 660 /var/run/docker.sock
jay@LAPTOP-HUF2I1SF:/mnt/c/Users/JAY/serverless-url-shortener$ ls -l /var/run/docker.sock
srw-rw---- 1 root docker 0 Apr 27 10:10 /var/run/docker.sock
```

```
jay@LAPTOP-HUF2I1SF:/mnt/c/Users/JAY/serverless-url-shortener$ aws sts get-caller-identity
{
  "UserId": "AIDAXCN5E22HI6KWMVBBO",
  "Account": "486264854158",
  "Arn": "arn:aws:iam::486264854158:user/JAY@012"
}
```

```
jay@LAPTOP-HUF2I1SF:/mnt/c/Users/JAY/serverless-url-shortener$ aws ecr create-repository --repository-name url-shortener-repo --region us-east-1
{
  "repository": {
    "repositoryArn": "arn:aws:ecr:us-east-1:486264854158:repository/url-shortener-repo",
    "registryId": "486264854158",
    "repositoryName": "url-shortener-repo",
    "repositoryUri": "486264854158.dkr.ecr.us-east-1.amazonaws.com/url-shortener-repo",
    "createdAt": "2026-04-27T10:19:56.998000+00:00",
    "imageTagMutability": "MUTABLE",
    "imageScanningConfiguration": {
      "scanOnPush": false
    },
    "encryptionConfiguration": {
      "encryptionType": "AES256"
    }
  }
}
```

```
jay@LAPTOP-HUF2I1SF:/mnt/c/Users/JAY/serverless-url-shortener$ aws ecr get-login-password --region us-east-1 | \
docker login --username AWS --password-stdin 486264854158.dkr.ecr.us-east-1.amazonaws.com
Login Succeeded
```

```
jay@LAPTOP-HUF2I1SF:/mnt/c/Users/JAY/serverless-url-shortener$ docker build -f docker/Dockerfile -t url-shortener-repo .
[+] Building 33.2s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 316B                                0.1s
=> [internal] load metadata for public.ecr.aws/lambda/python:3.11 4.8s
=> [internal] load .dockerignore                                   0.1s
=> => transferring context: 2B                                       0.1s
=> [1/5] FROM public.ecr.aws/lambda/python:3.11@sha256:2a18e270d1a4cdcf56930bb007e64044f9160957c743badf07f62c0cb3e75964 25.0s
```

```
jay@LAPTOP-HUF2I1SF:/mnt/c/Users/JAY/serverless-url-shortener$ docker tag url-shortener-repo:latest 486264854158.dkr.ecr.us-east-1.amazonaws.com/url-shortener-repo:latest
jay@LAPTOP-HUF2I1SF:/mnt/c/Users/JAY/serverless-url-shortener$ docker push 486264854158.dkr.ecr.us-east-1.amazonaws.com/url-shortener-repo:latest
The push refers to repository [486264854158.dkr.ecr.us-east-1.amazonaws.com/url-shortener-repo]
c55176c1a45e: Pushed
a1aaa884949b: Pushed
daf17606c1f6: Pushed
fdf5e8cffa33: Pushed
4b6d42e6844e: Pushed
72b43d245193: Pushed
626e8abb4e79: Pushed
685683a0c915: Pushed
fec0d3cb6efb: Pushed
89d695272527: Pushed
393064f630e5: Pushed
latest: digest: sha256:8157bd30c17442f7d460427112c8c5d9be749b470669a1b69254d9782ee8344b size: 856
```

```
jay@LAPTOP-HUF2I1SF:/mnt/c/Users/JAY/serverless-url-shortener$ chmod +x deploy.sh
```

```
./deploy.sh ap-south-1
```

```
=====
```

```
Deploying Serverless URL Shortener
```

```
Region:  ap-south-1
```

```
Account: 486264854158
```

```
=====
```

```
[1/4] Initializing Terraform...
```

```
Initializing the backend...
```

```
Initializing provider plugins...
```

- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v5.100.0

```
jay@LAPTOP-HUF2I1SF:/mnt/c/Users/JAY/serverless-url-shortener$ terraform -chdir=terraform apply
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```
# aws_api_gateway_deployment.url_shortener will be created
```

```
+ resource "aws_api_gateway_deployment" "url_shortener" {  
  + created_date = (known after apply)  
  + execution_arn = (known after apply)  
  + id           = (known after apply)  
  + invoke_url   = (known after apply)  
  + rest_api_id  = (known after apply)  
}
```